

Ejercicios de los viernes (14/08/2026)

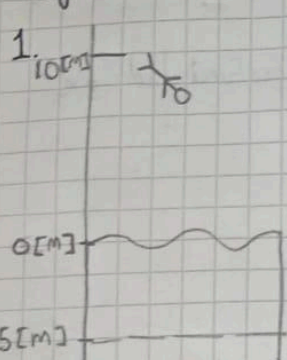
Materia: Mecánica Clásica

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1. a)

Ejercicios mecánica

1. 

— Para la situación plataforma — agua: — //

$$E_i = E_f \rightarrow \frac{1}{2} m v_0^2 + mgh = \frac{1}{2} m v_e^2 + mgh$$

$$h = 10[m]; v_e = ? \rightarrow 10mg = \frac{1}{2} m v_e^2$$

$$v_e = \sqrt{20g} \left[\frac{m}{s} \right]$$

— Para la situación agua — suelo — //

Newton: $\Sigma F = m\dot{v}$; $\Sigma F = mg - E - bv$, siendo E la fuerza de empuje y bv la fuerza viscosa con b el coeficiente de amortiguamiento viscoso. También, para un ser humano, podemos considerar $mg = E = pg \sqrt{v}$

Entonces: $-bv = m\dot{v} \rightarrow \frac{-b}{m} = \frac{dv}{dt} \frac{1}{v} \rightarrow \frac{-b}{m} dt = \frac{dv}{v}$

$$\rightarrow \frac{-b}{m} \int dt = \int \frac{dv}{v} \rightarrow \frac{-b}{m} t = \ln(v) + C \rightarrow v = e^{\frac{-b}{m}t + C}$$

$$\rightarrow v(t) = C e^{\frac{-b}{m}t} \rightarrow v(0) = \sqrt{20g} = C$$

$$\frac{dr}{dt} = \sqrt{20g} e^{\frac{-b}{m}t} \rightarrow \int_0^r dr = \sqrt{20g} \int_0^t e^{\frac{-b}{m}t} dt = \sqrt{20g} \left[-\frac{b}{m} e^{\frac{-b}{m}t} \right]_0^t =$$

$$= \sqrt{20g} \left(\frac{b}{m} - \frac{b}{m} e^{\frac{-b}{m}t} \right); \text{ si } t \rightarrow \infty \rightarrow r = \sqrt{20g} \left(\frac{m}{b} \right)$$

Para un humano promedio: $b = 200 \frac{kg}{s}$ y $m = 70 kg$

$$r = \sqrt{20g} \cdot \frac{70}{200} = 1.9 [m] \rightarrow (a)$$

1. b)

1.
b) Corcho esférico $r = 0.025 \text{ [m]}$

5 cm
0 cm

$\Sigma F = \frac{d(mv)}{dt}$. $\Sigma F = -mg + \rho g V - \frac{1}{2} \rho C_d A v^2$; $\Sigma F = m \frac{dv}{dt} = m \dot{v}$

$\rightarrow m \dot{v} = \rho g V - (mg + \frac{1}{2} \rho C_d A v^2) \rightarrow \frac{\rho g V - mg}{m} = c t e^t = \beta$

$\rightarrow \frac{1}{2} \rho C_d A = c t e = \alpha \rightarrow \frac{dv}{dt} = \beta - \alpha v^2 \rightarrow \frac{dv}{\beta - \alpha v^2} = dt$

$\int \frac{dv}{\beta - \alpha v^2} = t + C$; $\int \frac{dv}{\beta - \alpha v^2} = \frac{1}{2\sqrt{\alpha\beta}} \left(\ln \left| \frac{\sqrt{\beta} v + 1}{\sqrt{\beta} v - 1} \right| \right)$

$\sqrt{\frac{\beta}{\alpha}} = \sigma \rightarrow \int \frac{dv}{\beta - \alpha v^2} = \frac{1}{2\sqrt{\alpha\beta}} \left(\ln \left| \frac{\sigma v + 1}{\sigma v - 1} \right| \right) = t + C$

$\frac{\sigma v + 1}{\sigma v - 1} = c e^t \rightarrow \sigma v + 1 = c e^t \sigma v - c e^t \rightarrow \sigma v - c e^t \sigma v = -1 - c e^t$

$\frac{\sigma v + 1}{\sigma v - 1} = c e^t \rightarrow v(\sigma - \sigma c e^t) = -1 - c e^t \rightarrow v = \frac{-1 - c e^t}{\sigma - \sigma c e^t}$

1.
b) $\frac{1}{2\sqrt{\alpha\beta}} \left(\ln \left| \frac{\sigma v + 1}{\sigma v - 1} \right| \right) = t + C \rightarrow \ln \left| \frac{\sigma v + 1}{\sigma v - 1} \right| = 2\sqrt{\alpha\beta} (t + C)$

$\frac{\sigma v + 1}{\sigma v - 1} = C e^{2\sqrt{\alpha\beta} t} \rightarrow \sigma v + 1 = (\sigma C e^{2\sqrt{\alpha\beta} t} v - C e^{2\sqrt{\alpha\beta} t})$; $e^{2\sqrt{\alpha\beta} t} = f(t)$

$\rightarrow v(\sigma - \sigma C f(t)) = -(1 + C f(t))$

$\rightarrow v = -\frac{1}{\sigma} \left(\frac{1 + C f(t)}{1 - C f(t)} \right) \rightarrow v(0) = -\frac{1}{\sigma} \left(\frac{1 + C}{1 - C} \right) = 0$

$\rightarrow \frac{1 + C}{1 - C} = 0 \rightarrow 1 + C = 0 \rightarrow C = -1$

$v(t) = -\frac{1}{\sigma} \left(\frac{1 - f(t)}{1 + f(t)} \right)$; $\sigma = \sqrt{\frac{\beta}{\alpha}}$; $\frac{1}{2} \rho C_d A = \alpha$

$\alpha = \frac{1}{2} (1000) (0.5) (\pi \cdot 0.025^2) \approx 0.49 \left[\frac{\text{kg}}{\text{m}} \right]$

$\beta = \frac{\rho g V - mg}{m}$; $\rho_{\text{corcho}} = 240 \frac{\text{kg}}{\text{m}^3} \rightarrow V = \frac{4}{3} \pi (0.025)^3 = 6.5 \times 10^{-5}$

1. b)

$$\beta = \frac{\rho g V - m g}{m} ; \rho_{\text{torcho}} = 240 \frac{\text{kg}}{\text{m}^3} \rightarrow V = \frac{4}{3} \pi (0.025)^3 = 6.5 \times 10^{-5}$$

$$V \rho_{\text{torcho}} = m = 0.016 \text{ [Kg]} ; \beta = \frac{(1000)(9.8)(6.5 \times 10^{-5}) - (0.016)(9.8)}{(0.016)} = 30.0$$

$$\sigma = \sqrt{\frac{0.49}{30}} = 0.13 ; 2\sqrt{\alpha\beta} = 7.7$$

$$V(t) = -\frac{1}{0.13} \left(\frac{1 - e^{7.7t}}{1 + e^{7.7t}} \right) = -7.7 \left(\frac{1 - e^{7.7t}}{1 + e^{7.7t}} \right) = -7.7 \left(\frac{1}{1 + e^{7.7t}} - \frac{e^{7.7t}}{1 + e^{7.7t}} \right)$$

$$V(\infty) = 7.7 \left(\frac{e^{7.7t}}{1 + e^{7.7t}} - \frac{1}{1 + e^{7.7t}} \right) = 7.7$$

$$V_f = 7.7 \left[\frac{\text{m}}{\text{s}} \right]$$

2. a)

$$2. \vec{F} = e\vec{E} + \frac{e}{c} \vec{v} \times \vec{B}$$

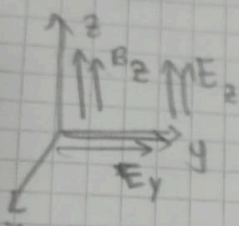
a) $|\vec{E}| = 0 ; \vec{F} \perp \vec{B} \rightarrow \vec{v} \perp \vec{B}$

$$\vec{F} = \frac{e}{c} \vec{v} \times \vec{B} = \frac{e}{c} |\vec{v}| |\vec{B}| \sin\left(\frac{\pi}{2}\right) \hat{n} = \frac{evB}{c} \hat{n}$$

$$\sum \vec{F} = m\vec{a} \rightarrow F = m\alpha ; \alpha = \frac{v^2}{r} \rightarrow F = \frac{mv^2}{r} = \frac{evB}{c} \rightarrow r = \frac{mcv}{eB}$$

2. b)

b) Tenemos



$$\rightarrow m \left(\frac{dv_x}{dt} + \frac{dv_y}{dt} + \frac{dv_z}{dt} \right) = e (0, E_y, E_z) + \frac{e}{c} \vec{v} \times (0, 0, B_z)$$

El producto cruz:

$$(v_x, v_y, v_z) \times (0, 0, B_z) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_x & v_y & v_z \\ 0 & 0 & B_z \end{vmatrix}$$

$$= \hat{i}(v_y B_z) - \hat{j}(v_x B_z) + \hat{k}(0)$$

Para $\hat{k}$: $m \frac{dv_z}{dt} = e E_z + 0$

$$\Rightarrow m \frac{dz}{dt} = \frac{e E_z}{m} dz \rightarrow \begin{cases} \dot{z}(t) = \frac{e E_z}{m} t + C_1 \\ z(t) = \frac{e E_z}{2m} t^2 + C_1 t + C_2 \end{cases}$$

Con P.V.I:

$$z(0) = z_0 \wedge \dot{z}(0) = \dot{z}_0 \Rightarrow z(0) = \frac{e E_z}{2m} (0)^2 + C_1 (0) + C_2 = z_0$$

$$z(t) = \frac{e E_z}{2m} t^2 + \dot{z}_0 t + z_0 \quad \leftarrow \dot{z}(0) = \frac{e E_z}{m} (0) + C_1 = \dot{z}_0$$

3. a)

3. Una gota de lluvia de masa inicial m_0 gramos, cae, partiendo del reposo, y cruza una nube cuyo espesor es de a [cm]. A medida que cae va ganando masa a razón de b [g/seg]. Las gotitas que forman la nube están en reposo con relación a la Tierra. El movimiento de la gota encuentra una resistencia proporcional a su velocidad.

m_0 [g]; $v_0 = 0$; espesor_nube = a [cm]; $m(t) = m_0 + bt$; $v_{nube} = 0$; $F_r = -Kv$

a) Plantear la ED.

Newton: $\Sigma F = \frac{d(mv)}{dt}$; $\Sigma F = mg - Kv = (m_0 + bt)g - Kv$

$\rightarrow (m_0 + bt)g - Kv = \frac{d(mv)}{dt}$; tanto $m(t)$ como $v(t)$ dependen del tiempo.

$\frac{d(mv)}{dt} = m \frac{dv}{dt} + v \frac{dm}{dt}$; $\frac{dm}{dt} = b$; $(m_0 + bt)g - Kv = m \frac{dv}{dt} + bv$

$\rightarrow m \frac{dv}{dt} = mg - v(b+K) \rightarrow mg = m \frac{dv}{dt} + v(b+K) \rightarrow g = \frac{dv}{dt} + v \left(\frac{b+K}{m} \right)$

$y_p = \dot{v} + v \left(\frac{b+K}{m} \right) = g$

$(p) \dot{v} + v \left(\frac{b+K}{m} \right) = g(p)$

$Q(t) = \frac{dx}{dt} + xP(t)$

$y_H = \frac{dv}{dt} = -v \left(\frac{b+K}{m} \right)$

$\frac{dv}{v} = - \left(\frac{b+K}{m} \right) dt$

$\ln v = - \left(\frac{b+K}{m} \right) t + C$

3. a) y b)

$$g = \frac{dv}{dt} + v \left(\frac{b+k}{m} \right) \rightarrow \text{E.D. Lineal de 1º orden} \rightarrow \frac{dv}{dt} + P(t)v = Q(t)$$

$$P(t) = \frac{b+k}{m}; Q(t) = g. \text{ El factor integrante: } \mu(t) = e^{\int P(t) dt}$$

★ Multiplicar a $\left[\frac{dv}{dt} + P(t)v \right]$ por cierta función especial convierte

$\left[\frac{dv}{dt} + P(t)v \right]$ en la derivada de un producto, esto es el factor integrante

$$\text{Queda: } \mu(t) \left[\frac{dv}{dt} + P(t)v \right] = \frac{d}{dt} (\mu \cdot v) = \mu Q; e^{\int P(t) dt} = e^{\int \frac{b+k}{m} dt} = e^{\frac{b+k}{m} t}$$

$$\left(\frac{1+k}{b} \right) \int \frac{b}{m_0+bt} dt = \ln(m_0+bt) \cdot \left(\frac{1+k}{b} \right) = \ln(m_0+bt) \left(\frac{b+k}{b} \right)$$

$$e^{\ln(m_0+bt) \left(\frac{b+k}{b} \right)} = (m_0+bt)^{\left(\frac{b+k}{b} \right)}; \frac{b+k}{b} = N \rightarrow \mu(t) = (m_0+bt)^N$$

$$\rightarrow \frac{d}{dt} [(m_0+bt)^N \cdot v] = (m_0+bt)^N \cdot g; \mu(t) \cdot v = \int \mu(t) Q(t) dt + C$$

$$\rightarrow (m_0+bt)^N \cdot v = \int (m_0+bt)^N \cdot g dt + C; u = m_0+bt; du = b dt \rightarrow dt = \frac{du}{b}$$

$$\rightarrow \frac{g}{b} \int u^N du; N \neq -1 \rightarrow \int u^N du = \frac{u^{N+1}}{N+1} = \frac{(m_0+bt)^{N+1}}{N+1} + C$$

$$\rightarrow (m_0+bt)^N \cdot v = \frac{g}{b} \frac{(m_0+bt)^{N+1}}{N+1} + C; N = \frac{b+k}{b} = 1 + \frac{k}{b}$$

$$v \cdot (m_0+bt)^{\left(\frac{b+k}{b} \right)} = \frac{g}{b} \frac{(m_0+bt)^{\frac{b+k}{b} + 1}}{\frac{b+k}{b} + 1} + C \rightarrow v(t) = \frac{g}{2b+k} (m_0+bt) + C(m_0+bt)^{-N}$$

$$v(0) = \frac{g}{2b+k} (m_0) + C m_0^{-N} = 0 \rightarrow C = -\frac{g}{2b+k} \frac{(m_0)}{(m_0)^N} = -\frac{g}{2b+k} m_0^{1-N}$$

$$v(t) = \frac{g}{2b+k} \left[\underbrace{m_0+bt}_m - m_0^{N+1} (m_0+bt)^{-N} \right]$$

en $x(t) = 0 \rightarrow m(0) = m_0$ en $x(t_a) = a \rightarrow m(t_a) = 2m_0 = m_0 + bt_a \rightarrow bt_a = m_0$

$$b) v(t_a) = \frac{g}{2b+k} \left[2m_0 + m_0^{N+1} (2m_0)^{-N} \right] = \frac{g}{2b+k} \left[2m_0 + m_0 2^{-N} \right] = \left| t_a = \frac{m_0}{b} \right.$$

$$= \frac{g m_0 (2 - 2^{-N})}{2b+k}$$

3. c)

c) gota

$$\sum F = \frac{dp}{dt} = m \frac{dv}{dt}; m = 2m_0 \rightarrow \sum F = m \frac{dv}{dt} = mg - Kv \rightarrow 2m_0 \frac{dv}{dt} = 2m_0 g - Kv$$

La vel. límite es donde $a = 0 = \frac{dv}{dt} \rightarrow \frac{dv}{dt} = g - \frac{Kv}{2m_0} = 0 \rightarrow \frac{Kv}{2m_0} = g \rightarrow$

$$\rightarrow V_{\lim} = \frac{2m_0 g}{K}$$